

Strategic Monitoring and Alignment to Achieve Business Process Best Practices

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Abstract

Business process design, reengineering and monitoring enable companies to achieve market-wide competitiveness and reduce their reaction time to environmental changes. IT systems become more sophisticated to meet the demands for real-time performance measurement and management. Besides, strategic management approaches such as the Balanced Scorecard gain more and more importance due to lacking alignment and strategic orientation in companies. Holistic management systems are therefore emerging. This paper focuses on a strategic management and monitoring framework that incorporates holistic aspects from the balanced scorecard with demands to data- and business driven process design and controlling. As a result, not a specific monitoring application is presented, but a methodology to combine strategic and tactical monitoring under best-practice assumptions.

1. Introduction

In the last years, many companies have focused on business process modeling and reengineering. To achieve competitiveness and to reduce market reaction times, business process management has become a significant element of day-to-day operations [11]. The automation of business processes execution and the integrated measurement of the results has been ongoing with this development. Performance management has risen to a popular management system. However, empirical studies showed poor results on it's holistic claim; e.g. too many financial indicators, lack of systematically measured process results etc. [12]. Focusing only on process-based – and in particular on workflow-based – monitoring and controlling, theoretical approaches have delivered profound research on setting up integrated cycles from process design to workflow implementation, execution and

controlling [14] to overcome these systematical and operational shortcomings.

The remainder of this article is intended to integrate the mentioned results into a strategic monitoring framework that is based on the assumptions of strategic, holistic management (Balanced Scorecard). A fundamental, business-driven guideline for business process modeling, execution and monitoring to achieve business process excellence and best practices is presented.

2. Holistic framework for strategic business (process) monitoring

Business monitoring approaches can be structured according to different criteria. A hierarchical classification is shown in Fig.1 that serves as basis for the identification of business monitoring levels.

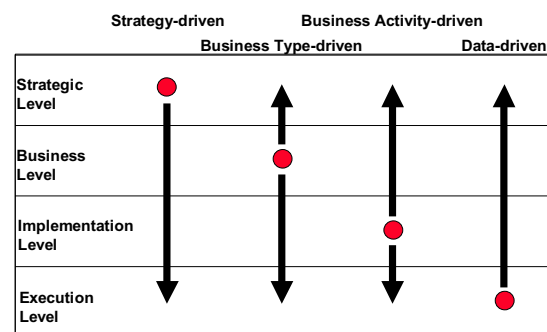


Figure 1: Business monitoring classifications [8]

Referring to existing approaches and architectures that cover business-activity and data-driven monitoring (e.g. corporate performance measurement [6] and workflow-based monitoring and controlling systems [15], [13], [5] as well as therefore designed service-oriented architectures [1]) it seems that mainly strategy-driven theories offer the possibility to a holistic integration into a company-wide, strategy-based management system, due to their higher level of

abstraction. Key elements of strategy-driven process design and monitoring are process maps that can be structured according to different criteria (relevance, time sequence, business or technical context, etc.) Even if widening this assumption and combining it with the findings of Kueng [11], that process performance measurement systems integrate aspects for identification of process goals and relevant indicators according to process strategies, a remarkable gap to company wide management systems (e.g. Balanced Scorecard) is obvious.

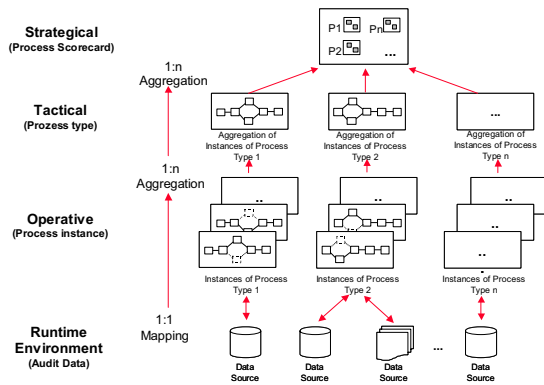


Figure 2: Business process aggregation levels [8]

Only a few papers can be found [5], that goals and indicators for different perspectives (e.g. market, finances and processes), interdependent by causal calculation-chains [3], are included in corporate performance management. The same idea is the basis for the Balanced Scorecard (BSC) concept [9], whereas the BSC has a strategic and not a process oriented starting point. Combining both aspects delivers a holistic framework for strategic business monitoring that does not only take quantitative but also qualitative goals into account. The Strategic Monitoring Framework (see Fig. 3) is supposed to be a top-down approach. Interactions and the integration into a learning process/loop shall be discussed in Chapter 3.

As referring to strategic alignment, claimed by the BSC methodology, the strategic level of this framework consists of a manageable set of goals and indicators that are directly derived from a company's vision and mission. Choosing a fairly equal number in each of four perspectives (finance, customers, processes and development focus) guarantees a holistic consideration of all success factors. Especially the following subjects are covered according to a strategic focus:

- Balanced performance monitoring on four business aspects
- Manageable number of performance indicators

- Integration of long-term goals with short term indicators
- Understandable cause-and-effect relationships between company-wide (performance) goals
- Consideration of quantitative and qualitative performance aspects
- Benchmarking possibilities due to strategic conformity

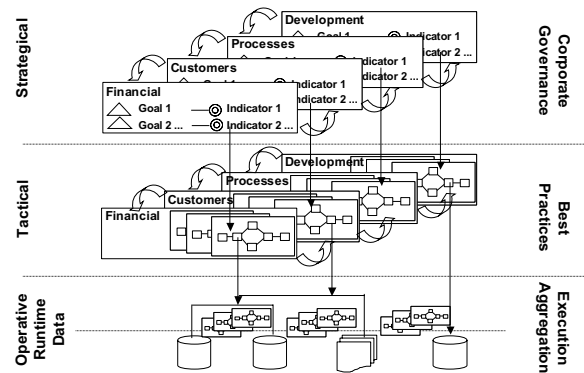


Figure 3: Strategic monitoring framework

The scorecarding concept on the strategic level of the framework guarantees design and optimization of the contents under a strategic focus. Tactic information is derived from the above-mentioned scorecard that will be structured again according to the selected perspectives. As depicted in figure 3, the tactic level is seen under the requirement of best practices. Unlike a typical drill-down procedure in business process monitoring [13] [15] that focuses on data-driven aspects, moving from the strategic to the tactic level consists in the detailing of processes (“zooming in”) according to their business relevance. Taking into consideration the holistic concept of this framework, the tactic level claims to represent “best practices”.

Depending on the strategic specifications “best practice” scenarios could consist of

- Comparisons between “should-be” and “as-is” processes
- Management guidelines
- Work instructions
- Legal preconditions (Commercial Code, US-GAAP rules, etc.)
- Business preconditions (ISO-certifications, etc.)

By moving forward from tactic to operative aspects, the main topics of process- and workflow-based monitoring (e.g. business to workflow graph transformation, audit data retrieval) are focused. The operative level is seen as an aggregation phase for data calculation based on audit and runtime data acquired

from the execution of processes. Due to the holistic approach of the strategic level, very heterogeneous data architecture must be expected at the operative level, including non-workflow orientated legacy systems and irregular, manual data inputs [12].

2.1. Tactical monitoring as business-driven drill down

Applying it to a business scenario can outline the added value of the presented framework. As the strategic level represents a selection of relevant goals and criteria, a closer view (“drill down”) is done by focusing on tactic information. As it is motivated to establish a comprehensive process driven monitoring and controlling system [5] [8], relevant information for indicators will be gathered by the integration of data-warehouse functionalities that are build up on runtime and audit data. This proceeding, as well as the obligatory transformation of business process paths [7] [4] to workflow/execution paths is presented in the left side of Figure 4. Multi-dimensional analytical capabilities are commonly referred to OLAP-systems in this context. The tactic level incorporates a further dimension, depicted as “business-driven” drill-down. As the user does not only receive status information about specific key performance indicators, their components and the actual (real-time monitoring) state of the system, further tactic information about the business parameters is added. This proceeding focuses on the design and reengineering capability as well as the plausibility of the executed business steps. The creation and application of business best practices is the aimed purpose.

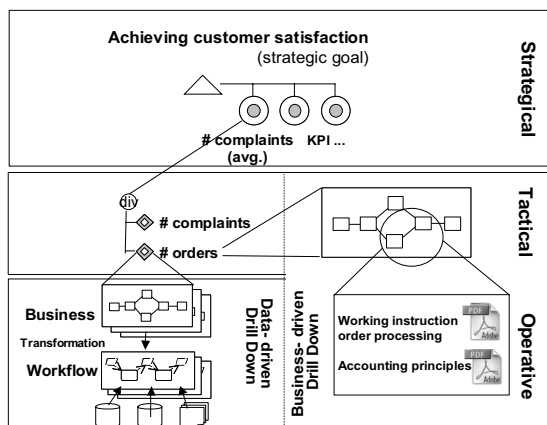


Figure 4: Typical framework application scenario

According to our experiences, the Sarbanes-Oxley (SOX) Act - specific financial and accounting disclosure requirements of mainly US-based

(Securities and Exchange Commission registered) companies - represents a typical application scenario, that demands not only business monitoring and reporting but further detailed tactic information about impacts and regulations posed by SOX (“how” and “why” of business acts) in the actual controlling process.

3. Feedback and learning process

As the previous chapter characterized a top-down approach for the Strategic Monitoring Framework (see Fig. 3), strategic alignment (as requested by BSC) has been realized by its application. Based on this assumption, an evaluation and learning process, displayed as looping process in Fig. 5, is proposed as well for the design and reengineering phase as for the monitoring and controlling aspects. The following graphic has been developed on the basis of the Business Process Management Systems (BPMS) Paradigm [10].

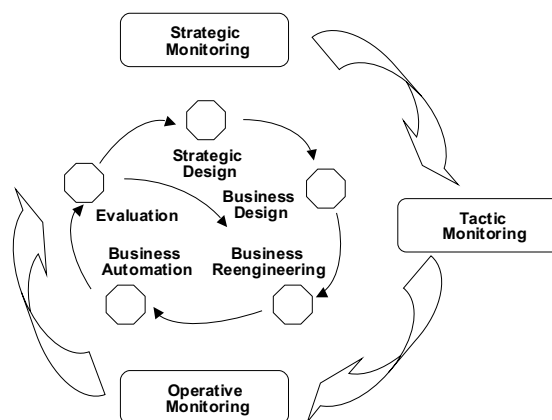


Figure 5: Monitoring framework as learning loop

Following the BMPMS Paradigm that is conventionalized as the inner circle, the system recommends a learning and evaluation loop under the demands of business best practices to be achieved by the application of the framework. Monitoring levels are sequential stages, requesting different information inputs and are directly constrained by the results of the previous stage. A closed system has been created by the integration of monitoring results into the evaluation stage in the design-loop.

Combining the findings of chapter 2 and 3 enables us to set up a monitoring framework that is focused on multi-dimensional (esp. data- and business-driven) controlling aspects under the premise of holistic, strategic alignment. The achievement of business

process excellence is a guideline, primarily identified to enable the integration of further - even “soft”-, not imperative process-oriented, information (e.g. legal requirements, instructions...).

4. IT integration of strategic design and monitoring

The application of this framework is a comprehensive challenge to IT systems due to its holistic character and the obvious need to integrate different application and data systems. For a case study that relates to service oriented architecture requested by this methodology, [1] must be stated.

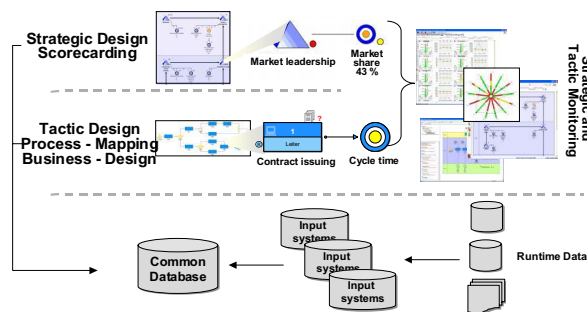


Figure 6: IT implementation of strategic and tactic monitoring

Strategic and tactical design is executed by applications, based on strong visual front-ends (online-mapping) and focuses on BSC and process mapping functionalities. Business driven information as well as data-driven aspects for indicator calculation are already designed. Interfaces to preliminary systems can be implemented online (database technologies, ODBC access) as well as on standardized exchange formats (XML). Aggregation and calculation of runtime data is performed by specialized workflow-based monitoring systems. The monitoring front-end for strategic and tactical information is represented by a web-based application, accessing as well design information and charts, as granting user-specific data access to runtime-monitoring systems. Apart from information visualization aspects, the front-end serves as analyzing and reporting application and represents a portal front end that combines detailed monitoring and business-specific information delivery (document management integration). Our approach has been implemented by using the business process management software ADONIS[®] and the Balanced Scorecard solution ADOScore[®].

5. Related work

Three main ideas have influenced this topic, which are the Balanced Scorecard methodology, business process design and reengineering and workflow technology.

The Balanced Scorecard has been developed in the 1990's by Drs. Kaplan and Norton [9] that still provide further research publications around the topic of strategic management. Whereas this topic has been more and more developed to a pure strategy-oriented, communication-driven approach that combines goals, indicators and projects, many business intelligence companies (e.g. SAS, Cognos, SAP) have tried to incorporate this theory into a corporate performance management component of their product portfolio that mainly consists of Data Warehouse technology. Most research work is done in specific topics that are part of the Balanced Scorecard. (e.g. financial, human resource management, process engineering, etc.)

As to the topic of business process design and reengineering, a lot of research has been performed upon the Business Process Management Paradigm [2] of the University of Vienna and the resulting software solution ADONIS. Further research work is based on findings of process-driven knowledge management.

Finally, workflow technology has been the enabler of business process automation. Whereas a lot of publications have appeared around interfaces of business process design and workflow transformation as well as the technical treatments of workflow automation according to the Workflow Management Coalition, less attention has been treated to more holistic aspects like (workflow-based) corporate performance management.

As [13] mentioned, systems like the HP Process Manager[®] (HPPM) or the ARIS[®] Process Performance Manager focus on distinctive analysis solutions based on process design, that can be coupled with workflow/runtime information of data warehouses by adaptors/interfaces and provide real-time process monitoring. It is obviously that these aspects are very technically/analytically oriented and lack a strategic methodological framework. In [1] and [5] elements of strategic oriented process monitoring can be found, whereas the focus is still set on a very operative (workflow execution) information.

6. Conclusion and future work

This paper presented a holistic framework, based on strategic management as starting point for corporate business performance management. The framework focuses not only on data-driven monitoring aspects, but enlarges the application scenario on tactical monitoring on business-driven information. Paying attention to these topics is seen to incorporate (strategic) guidelines and best practices into process management and design. Not process depending aspects will be monitored as requested by the strategic framework being set up on Balanced Scorecard preconditions.

Further work has to be done in the integration of Balanced Scorecard driven topics into the strategic monitoring framework and the identification of monitoring relevant aspects. Therefore research work in the area of business process execution (sense and response technologies) as well as knowledge-management and soft-skill monitoring shall be mentioned.

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